S2 Supplementary Data

Result-table index and reproducibility record

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Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S2.1 Frozen artefacts

The corrected occupancy/MNJ/L1/L4 table is `results/v1_task_20260922_d_eff_qc_excluded/task_part1.md`; its substantive BH family contains 45 pairwise tests and excludes the degenerate `reach_d_eff` readout while preserving that readout in a separate QC section. The L2/L3 panel is `results/v1_level_20260921/level_part1.md`; the chart-alpha duration diagnostic is `results/v1_duration_robustness_20260922/duration_robustness.md`; the L1 duration diagnostic is `results/v1_l1_duration_robustness_20260922/l1_duration_robustness.md`; the estimator-interaction diagnostic is `results/v1_estimator_task_interaction_20260922/interaction_report.md`; and the L4 certificate audit is `results/v1_l4_qc_20260921/l4_qc.md`. Adjacent JSON manifests preserve executable provenance. The recording inventory is `results/ingest_20260921/recording_table.csv`.

S2.2 Key complete-case contrasts

GG minus Rest	n	d_z	p	BH q
Chart-Jacobian alpha	44	1.42	1.558e-11	5.607e-10
One-step spectral abscissa	32	−0.328	0.1288	0.1783
L2 rotation rate	32	−1.10	6.906e-7	8.044e-6
L3 generator rotation	32	−1.08	6.906e-7	8.044e-6

Table 1: Rows transcribed from the separate L2/L3 family.

S2.3 Interpretation keys

The reported n is endpoint- and contrast-specific. Chart-Jacobian alpha is finite on 134/134 recordings, lag-1 quantities on 121/134, and history-conditioned L2 on 115/134. On the chart-alpha eligible 32-person GG–Rest subset, $d_z = 1.461$, median difference 0.01271, and $p = 3.26e-9$. The raw rows are not a formal test of effect-size difference; the separate standardized interaction test below is the declared exploratory test on common estimator support. The L2/L3 pairwise q column is global BH across 36 Wilcoxon tests.

S2.3b Standardized estimator-by-task interaction

The interaction runner uses the 121 recordings with both production-supported chart and one-step L3 alpha. Each endpoint is centered and scaled separately by its common-support pooled recording median and 1.4826 times MAD before subject-task medians and task contrasts are formed. The primary interaction is standardized chart contrast minus standardized L3 contrast. These three Wilcoxon tests have their own exploratory BH family; sign-flip and bootstrap columns are robustness summaries. This is normalization-dependent and does not compare raw endpoint units.

Contrast	n	median interaction	d_z	Wilcoxon p	BH q	sign-flip p
GG – Rest	32	1.888	0.966	6.79e-6	2.04e-5	1.00e-5
CC – Rest	32	0.555	0.327	0.0837	0.0837	0.0755
GG – CC	43	1.229	0.771	1.42e-5	2.13e-5	2.00e-5

Table 2: Exploratory estimator-by-task interaction on common production support. Positive values mean a larger standardized chart-alpha task contrast than standardized L3-alpha contrast.

S2.3c Common-support concordance

The concordance analysis uses the same complete subject pairs as the interaction output and introduces no new support or inferential family. Spearman ρ_s is a descriptive rank association between the standardized chart-alpha

and L3-alpha task contrasts. Sign agreement excludes zero contrasts from its denominator; bootstrap intervals resample complete subject pairs within each contrast.

Contrast	n	ρ_s	**95% CI**	p	**same sign**
GG – Rest	32	−0.383	[−0.683, −0.001]	0.0303	12/32
CC – Rest	32	−0.251	[−0.570, 0.099]	0.166	13/32
GG – CC	43	−0.284	[−0.559, 0.041]	0.0646	11/43

Table 3: Descriptive concordance of standardized estimator task contrasts. Bootstrap 95% CIs use 10,000 paired resamples. The axes are not raw endpoint units.

The source table, report and manifest are in `results/v1_estimator_concordance_20260922/`; the main-text scatter is `figures/results/ds004511_common_support_concordance_20260922.svg`.

S2.4 Complete L1 local-law family

The four frozen L1 metrics are shown for all three within-subject contrasts rather than selected by significance. The corrected q values use the 45-test substantive occupancy/MNJ/L1/L4-amplitude family; `reach_d_eff` is QC-only and is not part of that family. These are exploratory export-level summaries, not identified biological drift or diffusion.

Contrast	L1 metric	n	Δ med	d_z	p	BH q
GG – Rest	`diffusion_a_hat_frob`	44	0.03241	0.520	0.001358	0.003217
GG – Rest	`diffusion_total`	44	0.04987	0.551	0.001186	0.003139
GG – Rest	`diffusion_effective_dimension`	44	0.02645	0.146	0.3947	0.4674
GG – Rest	`chart_drift_b_hat_norm`	44	0.03436	1.82	2.16e-12	1.62e-11
CC – Rest	`diffusion_a_hat_frob`	45	0.008981	0.166	0.2715	0.3818
CC – Rest	`diffusion_total`	45	0.01813	0.155	0.3822	0.4674
CC – Rest	`diffusion_effective_dimension`	45	0.03825	0.0218	0.7122	0.7630
CC – Rest	`chart_drift_b_hat_norm`	45	0.005902	0.505	0.007377	0.01509
GG – CC	`diffusion_a_hat_frob`	44	0.02029	0.332	0.01170	0.02290
GG – CC	`diffusion_total`	44	0.02711	0.433	0.005605	0.01201
GG – CC	`diffusion_effective_dimension`	44	0.01941	0.182	0.3947	0.4674
GG – CC	`chart_drift_b_hat_norm`	44	0.02264	1.43	1.921e-11	1.081e-10

Table 4: Complete pairwise L1 family. Positive differences are later minus earlier as named in the contrast column. No row was selected by significance.

Complete-triplet Friedman tests likewise supported task variation in `diffusion_a_hat_frob` ($\chi^2 = 8.909$, $n = 44$, $q = 0.01938$), `diffusion_total` ($\chi^2 = 6.682$, $q = 0.04828$), and `chart_drift_b_hat_norm` ($\chi^2 = 49.95$, $q = 7.10 \times 10^{-}11$), but not `diffusion_effective_dimension` ($\chi^2 = 2.227$, $q = 0.3789$). These omnibus q values use the separate 15-test substantive Friedman family.